Korean Hate Speech Detection In Cross-lingual Transfer Learning Embedding Models

Team 26 | 20150839 Taegyeom Kim, 20170319 Kyungho Byoun, 이름, 20170499 Jeongjun Lee, 20224342 Minsun Kim

Introduction

In the online world, offensive language causes a great deal of conflict between people of different political ideologies, genders, races, and generations. And this is becoming a significant social problem in many nations. As a result, numerous NLP studies have been conducted to identify the various forms of offensive expressions, such as hate speech and cyberbullying. Recent research focuses heavily on offensive datasets in English, as there are numerous offensive datasets in English available on the Internet.

Numerous studies attempt to solve this issue by training the model using both English and the target language. This is referred to as the cross-lingual language model, or XLM for short. The advantage of XLM is that we can discover the model's latent space using a large number of English datasets. Then, we can fine-tune the model with a target language, such as Korean, to make it suitable for its intended purpose. Using this method, we can obtain a good performance from a model of the target language even if the source language is subpar.

Relative Works

BERT (Jacob Devlin et al., 2019) and its derivatives has shown novel improvements in most NLP tasks in recent. In hate speech, there are automatic hate speech detection approaches and BERT achieve 7-8% improvement compared to previous system (MacAvaney S et al., 2020). Kaggle competitions are actively held by big tech business such as Google and Meta and the leading model achieve about 80% accuracy in average. Recently, RoBERTa (Yinhan Liu et al., 2019) is frequently used and show high performance.

The evident challenge of studies on this topic is how to deal with different languages. Cross-lingual embedded models, XLM-R (Conneau et al., 2019), are suggested to take advantage of profoundness of English sources (Ranasinghe and Zampieri, 2020). It explain XLM-R are exploited to deal with non-English languages without performance degradation.

For Korean, there is single active Kaggle competition with Korean Hate Speech Dataset (Jihyung Moon et al., 2020). It suggests F1 score of benchmarks as a 56.9% and top of leaderboard has a about 66% accuracy. KoBERT, Multilingual BERT (Telmo Pires et al., 2019) are alternative approaches to deal with autonomous Korean hate speech detection. Relative position representation (Peter Shaw et al., 2018), korean specific tokenizer, data augmentation by GPT are additional way to improve the model.

Methods

Tackling difference between Korean and English

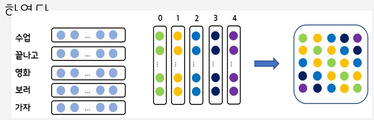
Cross-lingual transfer learning은 source language와 target language의 어순, POS와 같은 linguistic typology가 비슷한 언어에 더욱 효과적이라는 것이 알려졌다(Pires et al., 2019). 그리고 자연스럽게도 한국어와 영어는 차이를 보이는데, 한국어와 영어의 가장 큰 차이 중 우리가 해결하고자 집중한 것은 다음과 같다. (Matthew et al., 2013)

1) 영어는 한국어에 비해 어순을 비교적 엄격하게 요구하고 있다.

2) 한국어는 영어에 비해 다양한 어미를 학습하도록 요구하고 있다.

Cross-lingual transfer learning with relative position encoding

두 언어 모두 수식 관계 등 서로 연관 있는 단어는 가깝게 위치하는 경향이 있다. 이를 통해 단어의 절대적인 위치를 학습하는 것보단 단어의 상대적인 위치를 통해 특정 단어가 서로 가깝게 있음을 알리는 정보를 학습하는 것이 더 효과적일 것이라 가설을 세우고 XLM-R 모델에 **relative positional encoding을 적용**하여 각 단어의 상대적인 위치를 모델링 할 수 있도록 하였다.



Adapting morpheme tokenizer of Korean

한국어는 영어의 엄격한 어순 대신 추가적인 어미를 덧붙임으로써 단어 사이 관계를 모델링하기 때문에 영어만으로 이를 모델링하기엔 적합하지 않다. 따라서 영어에서 모델링 하지 못하는 한국어의 추가 정보를 명시적으로 모델링 해주어야 한다고 판단하였다. 우리는 이를 **형태소 토크나이저**를 사용하여 어미와 어간을 효과적으로 분리하고 이를 학습하고자 하였다.

[BERT Tokenizer] 수업 끝나고 영화 보러 가자 -> 수업 / 끝나고 / 영화 / 보러 / 가자

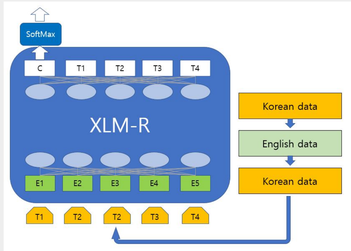
[Morpheme Tokenizer] 수업 끝나고 영화 보러 가자 -> 수업 / 끝나 / 고 / 영화 / 보 / 러 / 가 / 자

Data Augmentation: Random deletion, Word swapping

기존 한국어 데이터의 경우 약 7000개의 데이터가 라벨링되어 있다. 하지만, 데이터 augmentation을 통해 결과를 향상시킬 수 있을 것이라고 생각하고, 단어 랜덤 삭제와 단어 간 위치 교체를 통해 데이터를 50000개로 증가시켰다.

New Methodology: Sandwich pretraining

English-Korean-English순의 사전학습을 거친 후, Korean hate speech 학습을 수행하였다. 대량의 언어로 학습이 가능한 영어를 통해 영어와 한국어 사이의 관계성을 형성할 수 있을 것이라고 예상하였다.



Experiments

Datasets and Models

우리는 상대적으로 자료가 많은 영어를 source language로 설정, 한국어를 target language로 설정하였다. Pretraining을 위한 영어 hate speech dataset으로써 **Hate speech and Offensive Language dataset**을 사용, 한국어 hate speech dataset으로써 Korean Hate Speech dataset (Jihyung Moon et al., 2020)을 사용하였다. 특히 korean dataset은 Kaggle을 통해 성능 leaderboard를 제공해주었기 때문에 제안하는 모델 성능의 객관적 지표가 될 수 있을 것이라 판단하였다.

(data augmentation추가)

우리는 더 나아가서 Korean dataset의

제안하는 Model은 BERT 기반 모델인 XLM-R을 사용하였으며 모델 구조가 BERT와 매우 유사하기 때문에 Baseline모델을 BERT기반 모델로 설정하여 비교할 수 있었다.

Evaluation Metrics

정량적 지표로써 Accuracy와 F1 score를 사용하여 정확도를 모델링 할 수 있었다

Model comparison

1. Monolingual model vs Crosslingual model vs Multilingual model

   Crosslingual model이 기존의 monolingual model이나 multilingual model과 비교하여 좋은 성능을 내는지 확인하였다. Crosslingual model로써 Korean XLM-R model pretrained with English을 설정, Monolingual baseline model로써 KoBERT를 사용. Multilingual baseline model로써 multilingual BERT사용.

2. Absolute position encoding vs Relative position encoding

   우리는 영어와 한국어의 어순 차이를 유연하게 모델링 하기 위해 XLM-R 모델의 position encoding을 relative로 설정 후 실험하여 기존 absolute position encoding과의 성능을 비교하였다.

3. Wordpiece Tokenizer vs Morpheme Tokenizer

   한국어 토크나이저를 XLM-R model 의 기본 토크나이저인 Wordpiece Tokenizer를 형태소 토크나이저로 대체하여 학습. Pretraining을 위한 영어의 토크나이저는 Wordpiece Tokenizer로 고정하여 학습하였다.

4. Data augmentation vs Sandwich pretraining

   학습의 양을 늘리는 기법인 data augmentation 기법을 적용한 것과 우리가 제안하는 sandwich pretraining을 비교하였다.

Results

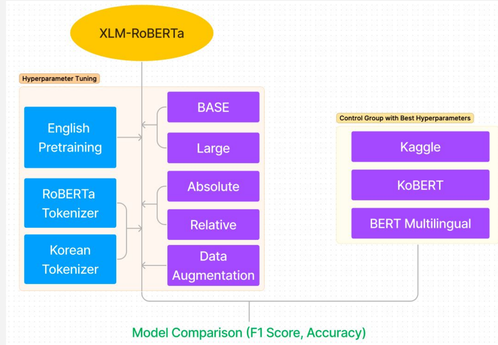


Figure 1. Experiment Model Division Mapping

Model	Evaluation Result	F1 Score macro/weighted
Kaggle Competition	0.677	unknown
BERT- Multilingual	0.548	
KoBERT	0.604	
XLM-R-base	0.650	0.641 / 0.651
XLM-R-base (relative position embedding)	0.555	0.533 / 0.550
XLM-R-base Eng-pretrain	0.618	0.612 / 0.622
XLM-R-base Eng-pretrain (relative position embedding)	0.606	0.593 / 0.609
XLM-R-large	0.669	0.667 / 0.662
XLM-R-large Eng-pretrain	0.671	0.667 / 0.675
XLM-R with self-trained tokenizer	0.652	0.647 / 0.643
Data Augmentation	0.639	0.632 / 0.622
Sandwich Pretraining	0.694	0.691 / 0.691
Sandwich Pretraining with self-trained tokenizer	0.690	0.688 / 0.685

Table1. Model features and Evaluation Results: For more details, please refer to the Figure 2.

우리는 한국어 혐오발언에 대해 서로 다른 모델과 전략에 따라 어떠한 결과가 도출되는지 알아보았다. 그 과정에서 각 모델의 특징과 더불어 동일한 데이터셋의 evaluation result를 알 수 있었다. 또한, F1 result를 통해 recall과 precision도 확인하였다.

먼저 각 모델의 특징에 대해 알아보았다. BERT의 경우, 대량의 코퍼스를 BERT에 사전학습시키고 분류를 원하는 데이터로 fine-tuning을 거쳐 모델을 완성한다. 특히 BERT-multilingual은 여러 언어들을 같이 학습하며 Masked language modeling을 통해 데이터 양이 부족한 언어에도 적용가능하도록 하였다. 이러한 BERT에 대량의 한국어 데이터로 사전학습된 BERT모델이다. XLM-RoBERTa역시 100개의 언어로 사전학습된 모델이다. xlm과 bert 의 가장 근본적인 차이는 parallel corpus이다. 번역문서에 대해서 단어 임베딩을 통해 단어사전을 생성한 후, CTK와 MLM을 통해 다중언어 코퍼스를 정렬한다. 우리는 XLM 모델의 positional embedding변경을 통해 어떤 상황에서 한국어에 더 잘 맞는 결과값이 나오는지 확인해보았다. 또한, 토크나이저 변경을 통해 어떠한 토크나이저가 한국어 성능이 가장 잘 나오는지도 분류하였다.

우리는 method에서 본 것과 같이 기존 XLM에서 추가적인 수정을 통해 혐오발언을 3가지 수위(혐오성없음, 공격적, 혐오적)로 나누어 판별을 진행하였다. 그 결과, [table]에서와 같은 결과가 나왔다. 기존 Kaggle competition leaderboard에서의 가장 좋은 성능을 보여준 것은 0.677이었는데, 기존의 BERT, KoBERT모델을 쓴 것보다 좋은 수치였다. XLM-absolute와 XLM-relative의 경우에서는 absolute을 사용하는 쪽이 한국어에서는 더 좋은 성능을 보여주었다. 또한, 기존의 XLM 토크나이저를 사용하는 것이, 다른 토크나이저를 embedding하는 것보다 더 나은 성능을 보여주었다. 가장 좋은 결과를 보여준 것은 sandwich pretraining 기법이다. 이 경우는 English-Korean-English를 차례로 사전학습하여 0.694의 최종 accuracy를 만들었다.

Conclusion

This work show that **XLM-R with cross-lingual transfer learning outperforms the other existing models in Korean hate speech detection**. We set Multi-lingual BERT and KoBERT which are the most frequently used models in Korean NLP tasks as a control group. Base on this result, it is now possible to achieve higher performance using additional English datasets and advanced researches directly on Korean data.

The results obtained by 10 different hyperparameters and its own assumptions. The best result is achieved by 'sandwich pretrain model' whose pipeline includes additional pretrain process with English hate speech dataset. It shows that about 4% improvement compare to naive XLM-R. We also supposed that relative positional encoding outperforms absolute one (Shaw et al., 2019) but there is no evident improvement. One interpretation is that it is caused by the difference between English and Korean.

To apply **Korean tokenizer** is the biggest challenge in this work. A linguistic awareness of given language makes a huge impact on performance specifically in Korean (Park et al., 2020). However, almost korean tokenizer have worse performance than RoBERTa tokenizer which is generalized tokenizer for XLM-R model. We expected that this results is by reason of different policy between tokenizers. In future work, it would be the first objective to fit existing Korean tokenizer such as Khaiii, Mecab into XLM-R model.

Future Works

- 관련 논문(Ranasinghe and Zampieri, 2020)에서 다른 타국어에 대한 결과값이 한국어에 비해 더 나은 결과를 보여주었다. 이 논문에서는 한국어를 다루지는 않았다. 하지만, 같은 방식을 따라 한국어를 분석하였을 때 한국어 결과는 예상보다 낮게 나왔다. 따라서 이는 한국어에 맞춰진 한국어 Specific 한 특성을 살려서 training을 수행한다면 조금 더 나은 결과를 보여줄 수 있을 것으로 보인다.
- 한국어 dataset과 조금 더 가까운 japanese dataset을 사전학습시킨다면 더 나은 결과를 얻을 수 있을 것이다. 이와 비슷하게 다른 더 데이터가 많은 언어를 사전학습 시킨다면 더 나은 결과를 얻을 수 있을 것이다.
- XLM-R XL model을 사용하는 경우 더 좋은 결과가 나오게 된다면 resouceful 한 언어로 데이터 부족한 언어 모델링 하는것이 성능 개선에 도움을 준다는 가정이 맞을 수 있을 것이다.
- Data augmentation을 기존 데이터 텍스트 편집을 이용하여 진행하였으나, GPT와 같은 large language model을 이용한 data augmentation을 수행한다면 더 나은 결과를 얻을 수 있을 것으로 보인다. 또한 번역 모델을 embedding하여 data augmentation을 진행하는 사례도 최근 찾아볼 수 있기 때문에 이를 이용한다면 더욱 도움이 될 것이다.

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번역 중 입니다

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Introduction

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Numerous studies attempt to solve this issue by training the model using both English and the target language. This is referred to as the cross-lingual language model, or XLM for short. The advantage of XLM is that we can discover the model's latent space using a large number of English datasets. Then, we can fine-tune the model with a target language, such as Korean, to make it suitable for its intended purpose. **Using this method, we can obtain a good performance from a model of the target language even if the source language is subpar. (이부분 이해가 명확하게 가지 않습니다!)**

Korean also has scarce datasets, so we apply cross-lingual language model to hate speech detection in Korean and compare the result from other existing hate-speech detection models.

Relative Works

BERT (Jacob Devlin et al., 2019) and its derivatives has shown novel improvements in most NLP tasks in recent. In hate speech, there are automatic hate speech detection approaches and BERT achieve 7-8% improvement compared to previous system (MacAvaney S et al., 2020). Kaggle competitions are actively held by big tech business such as Google and Meta and the leading model achieve about 80% accuracy in average. Recently, RoBERTa (Yinhan Liu et al., 2019) is frequently used and show high performance.

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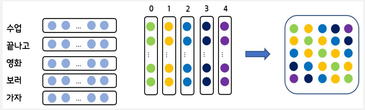
Methods

Tackling difference between Korean and English

Cross-lingual transfer learning is an effective way to learn a language with relatively little data through a source language with abundant data, but it is known that learning is more effective if source language and target language have similar linguistic typology such as word order and POS(Pires et al., 2019). And naturally, there are differences between Korean and English (Matthew et al., 2013), one of the main differences is that, in English, 'word order' is the most important factor of grammar, while, in Korean, 'end of a word' is important. In other words, **1) English requires a relatively strict word order compared to Korean**, while **2) Korean requires learning various endings(end of a word) compared to English**.

Cross-lingual transfer learning with relative position encoding

In both languages, words related to each other, such as modifying relationships, tend to be located closely. Through this, we hypothesized that it would be more effective to learn information indicating that certain words are close to each other through the relative position of the word rather than to learn the absolute position of the word. We apply **relative positional encoding** to the XLM-R model so that the relative position of each word could be modeled.



Adapting morpheme tokenizer of Korean

Also, as mentioned earlier, Korean models the relationship between words by adding additional endings(end of word) instead of strict word order in English. English data is not suitable to model the endings, an additional constraint on Korean. Therefore, it was judged that additional information in Korean that could not be modeled in English should be explicitly modeled. We expect that we will be able to effectively separate a word and its ending so that model them effectively by using **morpheme tokenizers**.

[BERT Tokenizer] 수업 끝나고 영화 보러 가자 -> 수업 / 끝나고 / 영화 / 보러 / 가자
[Morpheme Tokenizer] 수업 끝나고 영화 보러 가자 -> 수업 / 끝나 / 고 / 영화 / 보 / 러 / 가 / 자

Experiments

Datasets and Models

우리는 상대적으로 자료가 많은 영어를 source language로 설정, 한국어를 target language로 설정하였다. Pretraining을 위한 영어 hate speech dataset으로써 **Hate speech and Offensive Language dataset**을 사용, 한국어 hate speech dataset으로써 Korean Hate Speech dataset (Jihyung Moon et al., 2020)을 사용하였다. 특히 [2]은 Kaggle을 통해 성능 leaderboard를 제공해주었기 때문에 제안하는 모델 성능의 객관적 지표가 될 수 있을 것이라 판단하였다.

(data augmentation추가)

Korean dataset의

제안하는 Model은 BERT 기반 모델인 XLM-R을 사용하였으며 모델 구조가 BERT와 매우 유사하기 때문에 Baseline모델을 BERT기반 모델로 설정하여 비교할 수 있었다.

Evaluation Metrics

정량적 지표로써 Accuracy를 사용하였다. (F1 score 가능..?)

Model comparison

1. Monolingual model vs Crosslingual model vs Multilingual model

   Crosslingual model이 기존의 monolingual model이나 multilingual model과 비교하여 좋은 성능을 내는지 확인하였다. Crosslingual model로써 Korean XLM-R model pretrained with English을 설정, Monolingual baseline model로써 KoBERT를 사용. Multilingual baseline model로써 multilingual BERT사용.

2. Absolute position encoding vs Relative position encoding

   우리는 영어와 한국어의 어순 차이를 유연하게 모델링 하기 위해 XLM-R 모델의 position encoding을 relative로 설정 후 실험하여 기존 absolute position encoding과의 성능을 비교하였다..

3. Wordpiece Tokenizer vs Morpheme Tokenizer

   한국어 토크나이저를 XLM-R model 의 기본 토크나이저인 Wordpiece Tokenizer를 형태소 토크나이저로 대체하여 학습. Pretraining을 위한 영어의 토크나이저는 Wordpiece Tokenizer로 고정하여 학습하였다.

Results

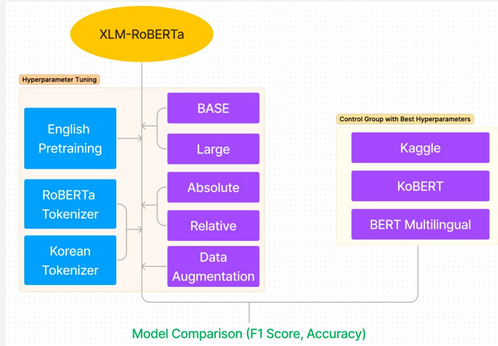


Figure 1. Experiment Model Division Mapping

Model	Evaluation Result	F1 Score macro/weighted
Kaggle Competition	0.677	unknown
BERT- Multilingual	0.548	unknown
KoBERT	0.604	unknown
XLM-R-base	0.650	0.641 / 0.651
XLM-R-base (relative position embedding)	0.555	0.533 / 0.550
XLM-R-base Eng-pretrain	0.618	0.612 / 0.622
XLM-R-base Eng-pretrain (relative position embedding)	0.606	0.593 / 0.609
XLM-R-large		
XLM-R-large Eng-pretrain	0.671	
XLM-R with self-trained tokenizer		
Data Augmentation		
Sandwich Pretraining	0.694	0.691/0.691
Sandwich Pretraining with self-trained tokenizer	0.690	0.688/0.685

Table1. Model features and Evaluation Results: For more details, please refer to the Figure 2.

We investigated the results of different models and strategies for hate speech in Korean. In the process, it was possible to know the evaluation result of the same dataset as well as the characteristics of each model. In addition, recall and precision were also confirmed through the F1 result.

In the case of BERT, a large corpus is pre-trained with BERT, and the model is completed through fine-tuning with the data to be classified. In particular, BERT-multilingual learns several languages at the same time, and through masked language modeling, it can be applied to languages with insufficient data. This BERT model is pre-trained with a large amount of Korean data. XLM-RoBERTa is also a pre-trained model in 100 languages. The most fundamental difference between xlm and bert is parallel corpus. After creating a word dictionary through word embedding for the translated document, the multilingual corpus is sorted through CTK and MLM. We checked under what circumstances the result values better suited to Korean came out through the positional embedding change of the XLM model. In addition, by changing the tokenizer, which tokenizer has the best performance in Korean is also classified.

As we saw in the method, we further modified the existing XLM-R to classify hate speech into three levels (non-hateful, aggressive, and hateful). As a result, the same result as in table 1 was obtained. The best performance on the existing Kaggle competition leaderboard was 0.677, which was better than using the existing BERT and KoBERT models. In the case of XLM-absolute and XLM-relative, using absolute showed better performance in Korean. Also, using the existing XLM-R tokenizer showed better performance than embedding other tokenizers. The sandwich pretraining technique showed the best results. In this case, English-Korean-English were pre-learned sequentially, resulting in a final accuracy of 0.694.

Conclusion

This work show that XLM-R with cross-lingual transfer learning outperforms the other existing models in Korean hate speech detection. We set Multi-lingual BERT and KoBERT which are the most frequently used models in Korean NLP tasks as a control group. Base on this result, it is now possible to achieve higher performance using additional English datasets and advanced researches directly on Korean data.

The results obtained by 10 different hyperparameters and its own assumptions. The best result is achieved by 'sandwich pretrain model' whose pipeline includes additional pretrain process with English hate speech dataset. It shows that about 4% improvement compare to naive XLM-R. We also supposed that relative positional encoding outperforms absolute one (Shaw et al., 2019) but there is no evident improvement. One interpretation is that it is caused by the difference between English and Korean.

To apply Korean tokenizer is the biggest challenge in this work. A linguistic awareness of given language makes a huge impact on performance specifically in Korean (Park et al., 2020). However, almost korean tokenizer have worse performance than RoBERTa tokenizer which is generalized tokenizer for XLM-R model. We expected that this results is by reason of different policy between tokenizers. In future work, it would be the first objective to fit existing Korean tokenizer such as Khaiii, Mecab into XLM-R model.

Future Works

- In the related paper (Ranasinghe and Zampieri, 2020), the results for other languages showed better results than Korean. This paper does not cover Korean. However, when analyzing Korean according to the same method, the Korean result was lower than expected. Therefore, it seems that if training is performed by utilizing the Korean specific characteristics tailored to the Korean language, it will be able to show better results.
- If the model is pre-trained with Japanese dataset, which is closer to the Korean, you will get better results. Similarly, better results could be achieved if it is pre-trained with another language which has more abundant data.
- If the XLM-R-XL model produces better results, it may be right to assume that modeling a data-poor language using data-rich language datasets helps improve performance.
- Data augmentation has been conducted by editing existing data text, but it seems that better results can be obtained by performing data augmentation using a large language model such as GPT. In addition, there is a recent case of data augmentation by embedding a translation model, so it would be more helpful to use it.

References

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Korean Hate Speech Detection In Cross-lingual Transfer Learning Embedding Models

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Problems of Existing Korean Hate Speech Models

- Recent research focuses heavily on offensive datasets in English, as there are numerous offensive datasets in English available on the Internet.
- Korean models the relationship between words by adding additional endings(end of word) instead of strict word order in English.
- In the paper, XLM-RoBERTa are exploited to deal with non-English languages without performance degradation, but there is clear degradation in Korean.

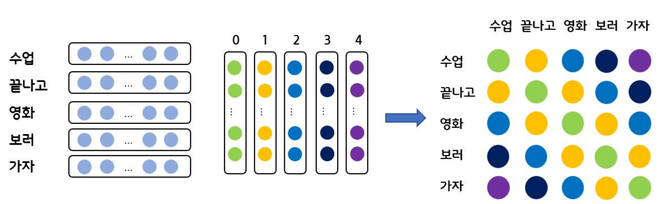
Goal of Our Work

1. Clearly measure the result of various models with one single dataset.
2. Improve the evaluation result better than the kaggle Korean hate speech competition winner.
3. Compare the performance between relative position encoding and absolute position encoding.
4. Using hate-speech specified Korean tokenizer to get better classification performance.
5. Propose our new training methodology, **Sandwich pretraining**, to get the better evaluation.

Methods

1. Cross-lingual transfer learning with relative position encoding

We hypothesized that it would be more effective to learn information indicating that certain words are close to each other through the relative position of the word rather than to learn the absolute position of the word.



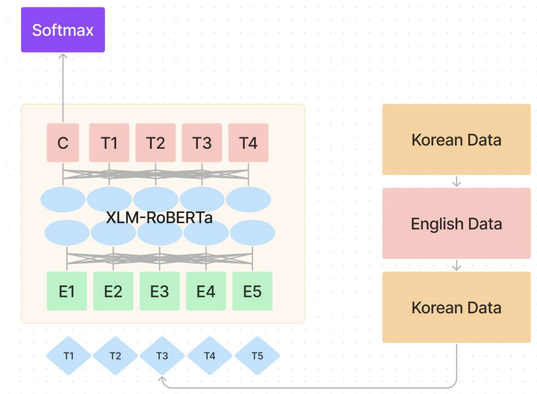
2. Adapting for Korean: Self-trained tokenizer

Basically, the XLM-R model uses RoBERTa tokenizer pretrained with 100 different languages for its generalization. However, there are some research claim that pretrained tokenizer for the specific language is better for good tokenization, especially in Korean. We expect that we will be able to effectively separate a word and its ending so that model them effectively by training tokenizer with Korean Hate speech dataset.

3. Data Augmentation: Random deletion, Word swapping

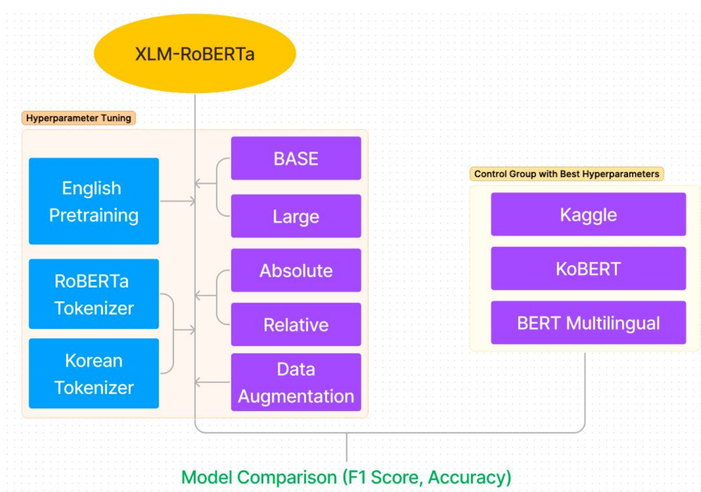
Thinking that the results could be improved through data augmentation, we increased the data about 7000 to 50,000 by randomly deleting words and replacing the positions between words.

4. New Methodology: Sandwich pretraining



Korean hate speech learning was performed after going through the preliminary learning in the order of Korean-English-Korean. It was expected that a relationship between English and Korean could be leveraged through this sandwich training method, which can be adapted to a variety of languages.

Results



Overall Experiment Mapping

Model	Accuracy	F1 Score
Kaggle Competition	0.677	unknown
BERT- Multilingual	0.548	unknown
KoBERT	0.604	unknown
XLM-R-base	0.65	0.651
XLM-R-large	0.669	0.662
Sandwich Pretraining	**0.694**	**0.691**

Table 1. Compare with Best Results From Existing Models

Model		Accuracy	F1 Score
XLM-R-base	Relative	0.555	0.550
	Eng-pretrain	0.618	0.622
	Eng-pretrain + Relative	0.606	0.609
XLM-R-large	Eng-pretrain	0.671	0.675
	Self-trained Tokenizer	0.652	0.643
	Data Augmentation	0.639	0.622
	Sandwich	**0.694**	**0.691**
	Sandwich + Self-trained Tokenizer	0.690	0.685

Table 2. Our Experiments Results

- Among the existing models, Kaggle competition result gets best accuracy.
- In the case of XLM-R-base, absolute encoding model with English pretraining gets best result.
- In the case of XLM-R-Large, sandwich pretraining method gets best accuracy.

Conclusion & Future Works

- This work show that **XLM-R with cross-lingual transfer learning outperforms the other existing models in Korean hate speech detection**. Base on this result, it is now possible to achieve higher performance using additional English datasets and advanced researches directly on Korean data.
- The **best result is achieved by 'sandwich pretrain model'** whose pipeline includes additional pretrain process with English hate speech dataset.
- We also supposed that **relative positional encoding outperforms absolute one but there is no evident improvement**. One interpretation is that it is caused by the difference between English and Korean.
- To apply Korean tokenizer is the biggest challenge in this work. A linguistic awareness of given language makes a huge impact on performance specifically in Korean. However, **self-trained tokenizer have worse performance than RoBERTa tokenizer** which is generalized tokenizer for XLM-R model.
- In future work, If we pre-train Japanese dataset, which is closer to Korean dataset than English, we will get better results. Similarly, pretraining with other data-rich languages will yield better results.